

Geophysics 3A – Week 3 Practical

Linear and Non-linear Inversion in Practice

Potential Fields & Geothermics

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Overview

This practical consolidates three pillars of inversion introduced in the Week 3 materials: (i) linear least squares (LLSQ) and weighted LLSQ; (ii) linearization of a nonlinear forward model (geometric mean of conductivity) to obtain a linear inverse problem; and (iii) Newton/ Gauss–Newton inversion for a nonlinear thermal conductivity model. You will:

1. build and fit a linear model between density and seismic velocity (Problem 1);
2. derive a linearized model for effective thermal conductivity from normative mineralogy and invert for mineral conductivities (Problem 2);
3. implement Newton’s method to fit laboratory thermal conductivity data for a plagioclase solid solution (Problem 3).

These tasks mirror the lecture/workshop derivations and the Week 2 computations, but now in Python with an emphasis on reproducible workflows.

Learning outcomes

By the end, you should be able to (i) formulate $\mathbf{d} = \mathbf{G}\mathbf{m}$ for linear/linearized problems; (ii) apply weighting and compute parameter uncertainties; (iii) construct a Fréchet/Jacobian matrix for a chosen nonlinear model and run damped Gauss–Newton updates; and (iv) articulate the tradeoff between misfit and regularization.

Data and file names

Place the following CSV files in the same folder as the notebook (exact column names may differ depending on the provided datasets):

- `density_velocity.csv` with columns such as RHO [kgm^{-3}] and VP [kms^{-1}].
- `thermo_mineralogy.csv` with columns: sample ID; measured effective thermal conductivity k_{eff} [$\text{Wm}^{-1}\text{K}^{-1}$]; normative mineral fractions w_{ji} for minerals $i = 1, \dots, p$ that sum to 1 for each sample j .
- `tc_plag.csv` with columns: molar fraction An (x), temperature T [K], conductivity k [$\text{Wm}^{-1}\text{K}^{-1}$].
- (Optional for comparison) `mineral_k_ref.csv` with columns: mineral name, k [$\text{Wm}^{-1}\text{K}^{-1}$] at reference T .

Submission

Submit a single PDF (export of the completed notebook with outputs) together with your Python notebook `week3_practical.ipynb`. Ensure figures are labeled and code is commented. Include brief answers to questions inline as Markdown cells.

1 Problem 1: Linear modeling of density vs. seismic velocity

You will fit a linear model relating density and V_P , extending Week 2.

Model forms. Consider the linear (in parameters) models:

$$\rho = m_0 + m_1 V_P, \quad (1)$$

$$\log \rho = n_0 + n_1 V_P, \quad (\text{alternative if a log-linear trend fits better}) \quad (2)$$

where parameters (m_0, m_1) or (n_0, n_1) are estimated by LLSQ.

Tasks.

1. Load the dataset and construct the operator matrix $\mathbf{A}$ (columns of ones and V_P). Solve $\min_{\mathbf{m}} \|\mathbf{d} - \mathbf{A}\mathbf{m}\|_2$.
2. Report best-fit parameters, 1σ uncertainties (from the parameter covariance), RMS misfit, and a residuals plot.
3. Repeat in log-space if warranted. Compare models visually and statistically; interpret in terms of lithology and scatter sources.
4. (Optional) Include heteroscedastic weights if uncertainties vary per datum.

2 Problem 2: Geometric mean conductivity from normative mineralogy (linearization)

Let a rock's effective thermal conductivity be modeled by a geometric mean of mineral conductivities

$$k_{\text{eff},j} = \prod_{i=1}^p k_i^{w_{ji}}, \quad \sum_{i=1}^p w_{ji} = 1, \quad (3)$$

for sample $j = 1, \dots, N$. Taking logarithms gives a linear system in the unknown mineral conductivities:

$$\log k_{\text{eff},j} = \sum_{i=1}^p w_{ji} \log k_i. \quad (4)$$

Define $d_j = \log k_{\text{eff},j}$, unknowns $m_i = \log k_i$, and $A_{ji} = w_{ji}$. Then $\mathbf{d} = \mathbf{A}\mathbf{m}$.

Tasks.

1. Assemble $\mathbf{A}$ from normative mineral fractions and $\mathbf{d}$ from measurements. Solve the LLSQ problem for $\mathbf{m}$ and report $k_i = e^{m_i}$ with 95% CIs.
2. Diagnose conditioning of $\mathbf{A}$ (rank, condition number). If ill-posed, add a small Tikhonov term $\lambda^2 \|\mathbf{m} - \mathbf{m}_0\|_2^2$ and comment on the trade-off.
3. Compare the inferred k_i with laboratory references when available (provide a table/plot if you have `mineral_k_ref.csv`). Discuss agreement and potential biases (texture, porosity, temperature).

3 Problem 3: Newton / Gauss-Newton inversion for a nonlinear conductivity model

Use the (nonlinear) plagioclase solid solution model

$$k(x, T) = (m_0 + m_1 x + m_2 x^2) \left(\frac{298}{T} \right)^{m_3}, \quad (5)$$

where x is molar fraction An and T is temperature in K. The Fréchet (Jacobian) entries are

$$\frac{\partial k}{\partial m_0} = \left(\frac{298}{T}\right)^{m_3}, \quad \frac{\partial k}{\partial m_1} = x \left(\frac{298}{T}\right)^{m_3}, \quad \frac{\partial k}{\partial m_2} = x^2 \left(\frac{298}{T}\right)^{m_3}, \quad (6)$$

$$\frac{\partial k}{\partial m_3} = (m_0 + m_1 x + m_2 x^2) \left(\frac{298}{T}\right)^{m_3} \log\left(\frac{298}{T}\right). \quad (7)$$

Tasks.

1. Implement a Gauss–Newton solver minimizing $\phi(\mathbf{m}) = \|\mathbf{d} - \mathbf{f}(\mathbf{m})\|_2^2 + \lambda^2 \|\mathbf{m} - \mathbf{m}_0\|_2^2$ with optional damping/line search. Start from a reasonable $\mathbf{m}_0$.
2. Report the parameter estimates with 95% confidence intervals from the approximate covariance $(\mathbf{F}^T \mathbf{F})^{-1}$ scaled by the residual variance.
3. Explore sensitivity to starting model and to λ ; comment on convergence/divergence and stability.
4. Plot fitted curves at several An values and inspect residuals vs. temperature.

Deliverables

For each problem include: figures (data + model; residuals), parameter tables with uncertainties, and brief interpretations (2–4 sentences each). Comment on linear vs. nonlinear structure, conditioning, and physical plausibility.

Appendix: Useful formulas

LLSQ and covariance. For $\mathbf{d} = \mathbf{A}\mathbf{m} + \boldsymbol{\epsilon}$, the LLSQ solution is $\hat{\mathbf{m}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{d}$. If the residual variance is $\hat{\sigma}^2 = \|\mathbf{d} - \mathbf{A}\hat{\mathbf{m}}\|_2^2 / \max(N - p, 1)$, then $\text{Cov}(\hat{\mathbf{m}}) \approx \hat{\sigma}^2 (\mathbf{A}^T \mathbf{A})^{-1}$.

Weighted LLSQ. With data weights $\mathbf{W}_d$, solve $\min \|\mathbf{W}_d(\mathbf{A}\mathbf{m} - \mathbf{d})\|_2$. Equivalent normal equations: $(\mathbf{A}^T \mathbf{W}_d^T \mathbf{W}_d \mathbf{A}) \mathbf{m} = \mathbf{A}^T \mathbf{W}_d^T \mathbf{W}_d \mathbf{d}$.

Geometric mean linearization. From Eq. (3), taking logs yields Eq. (4), linear in $\log k_i$.

Gauss–Newton step. With Jacobian $\mathbf{F}$, solve $(\mathbf{F}^T \mathbf{F} + \lambda^2 \mathbf{I}) \Delta \mathbf{m} = \mathbf{F}^T (\mathbf{d} - \mathbf{f}(\mathbf{m}))$ and update $\mathbf{m} \leftarrow \mathbf{m} + \alpha \Delta \mathbf{m}$ with optional step length α .

Notes. Use SI units consistently. If any dataset lacks uncertainties, assume equal weights or motivate a reasonable weighting scheme. Be explicit about any transformations (e.g., log space).